Lab Mid

CSE 4308

DATABASE MANAGEMENT SYSTEMS LAB

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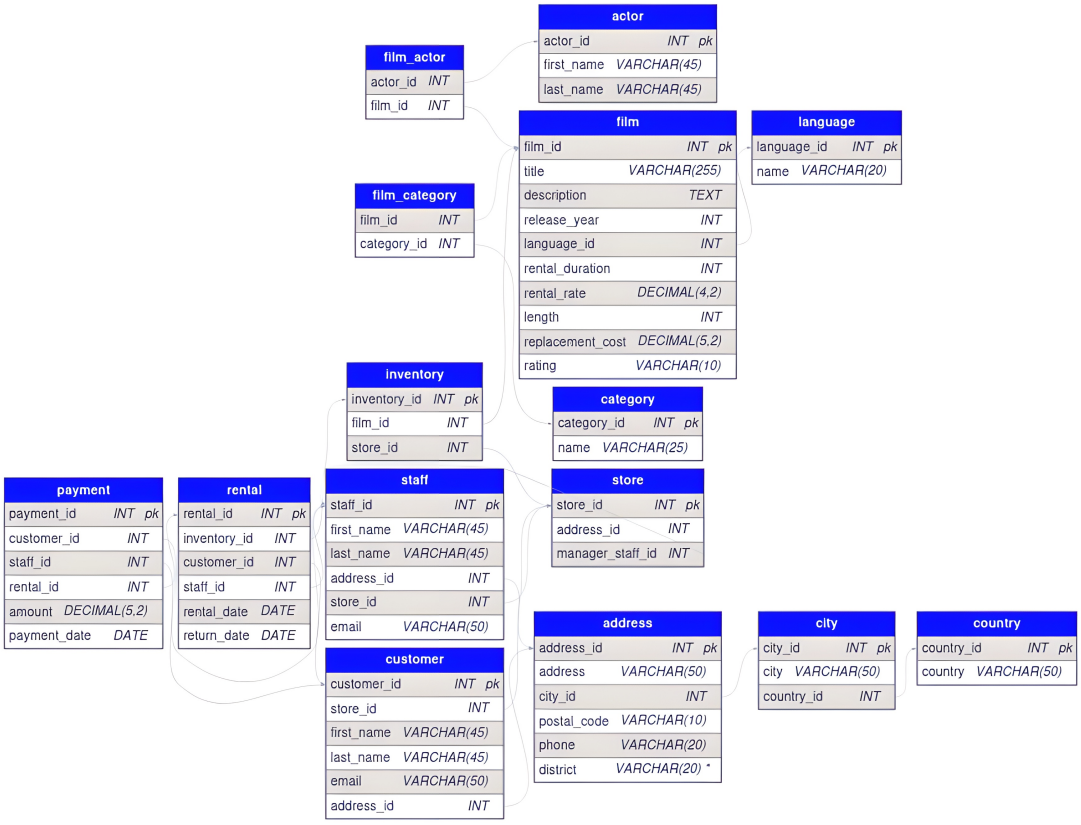
DBMS Lab Midterm



1. List all customers whose email contains the domain '.net'.

```
SELECT first_name, last_name, email
FROM customer
WHERE email LIKE '%.net%';
```

2. Find the titles of films that have ratings G or PG.

```
SELECT title
FROM film
WHERE rating IN ('G', 'PG');
```

3. Identify all films whose titles contain exactly two words.

```
SELECT title
FROM film
WHERE LENGTH(TRIM(title)) - LENGTH(REPLACE(TRIM(title), ' ', '')) = 1;
```

4. Show the number of customers who have made payments and who have never made a payment.

```sql
SELECT
(SELECT COUNT(DISTINCT customer_id) FROM payment) AS paid,
(SELECT COUNT(*)
 FROM customer
 WHERE customer_id NOT IN (
     SELECT customer_id FROM payment
 )) AS unpaid;
```

5. Display each store ID along with the total number of customers registered in that store.

```sql
SELECT store_id, COUNT(*) AS total_customers
FROM customer
GROUP BY store_id;
```

6. Find the film categories whose average rental rate is greater than 3.

```sql
SELECT c.name, AVG(f.rental_rate) AS avg_rate
FROM category c
JOIN film_category fc ON c.category_id = fc.category_id
JOIN film f ON fc.film_id = f.film_id
GROUP BY c.name
HAVING AVG(f.rental_rate) > 3;
```

7. List customers who have rented at least one film with a rental rate above 4.

```sql
SELECT DISTINCT c.customer_id, c.first_name, c.last_name
FROM customer c
WHERE c.customer_id IN (
    SELECT r.customer_id
    FROM rental r
    JOIN inventory i ON r.inventory_id = i.inventory_id
    JOIN film f ON i.film_id = f.film_id
    WHERE f.rental_rate > 4
);
```

8. Show rentals where the return date was exactly the same as the rental date.

```sql
SELECT rental_id, rental_date, return_date
FROM rental
WHERE EXTRACT(DAY FROM (return_date - rental_date)) = 0;
```

9. Identify customers who have rented more films than the average number of rentals per customer.

```sql
SELECT c.customer_id, c.first_name, c.last_name
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name
HAVING COUNT(r.rental_id) > (
```

```
        SELECT AVG(rental_count)
        FROM (
            SELECT COUNT(*) AS rental_count
            FROM rental
            GROUP BY customer_id
        ) sub
    );
```

10. Find films that have been rented more times than the average number of rentals per film.

```
    SELECT f.film_id, f.title, COUNT(r.rental_id) AS rental_count
    FROM film f
    JOIN inventory i ON f.film_id = i.film_id
    JOIN rental r ON i.inventory_id = r.inventory_id
    GROUP BY f.film_id, f.title
    HAVING COUNT(r.rental_id) > (
        SELECT AVG(film_rental_count)
        FROM (
            SELECT COUNT(r2.rental_id) AS film_rental_count
            FROM inventory i2
            JOIN rental r2 ON i2.inventory_id = r2.inventory_id
            GROUP BY i2.film_id
        ) sub
    );
```